Message Passing GNN for Antibiotic Discovery

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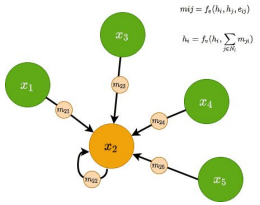
Based on *Deep learning-guided discovery of an antibiotic targeting Acinetobacter baumannii* by Liu et al. 2023

Introduction

We implemented '*Deep learning-guided discovery of an antibiotic targeting Acinetobacter baumannii*' (Nature, 2023) from Liu et al. This paper trains a message passing neural network to predict which molecules may display activity against A. Buamannii. A. Buamannii is a bacteria that causes severe healthcare associated infections.

We then ran their model on the same data to test their quantitative claims on the number of molecules the model predicts as being valid antibiotics, as well as their accuracy on the test set.

Additionally, we perform three ablation studies. We investigate whether the paper's use of an ensemble of ten models, a message passing depth of three, and zero dropout is the best choice they could have made. We will view these decisions through the frame of accuracy and compute.



Message Passing GNN, Source: *AISUMMER* 'Best GNN architectures'

A. Buamannii, Source: *MIT News*

Discussion

One thing we noted was that the paper didnt have that many data entires in its training dataset. This is likely because they had to develop the molecules in a lab, which the paper discusses in the Methods Section.

One shortcoming with our model and the dataset is that some predictions have a 100% certainty that they will inhibit the growth or kill the bacteria. This seems to be the result for compunds containing a Diazo/Diazonium group. This is an out of distribution prediction, as the papers original dataset did not contain any molecules with a Diazo/Diazonium Group.

Future work could include training on a broader dataset to ensure a more complete set of predictions, and eliminate the chance of the model having out of distribution predictions.

Methodology

We tested the model across ensemble size, dropout, and message passing depth and compared the AUROC across these parameter values.

We also ran inference on the COCONUT marine natural products dataset. This has over 500,000 molecules, all with a different chemical structure than the paper's Drug Repurposing Hub 5 molecules. The results of this inference were insightful into the model's ability to generalize for compounds with atomic features that are absent from the training set.

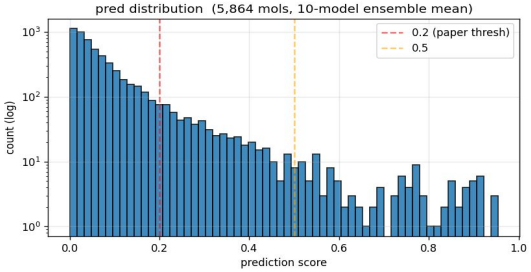
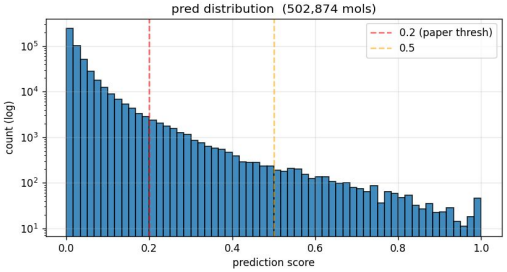
Results

Dropout raised AUROC above 1 sigma at 0.2 and 0.3.

Message passing depth made no significant difference on AUROC

With a single model, we matched the paper's 240 valid molecules. This match was a *coincidence* of the seed-0 split, not a true reproduction. Of our 240 priority molecules, only 148 (62%) survive the 10 model ensemble filter. This suggests that the paper's 240 valid molecules is not stable to model instantiation.

Additionally, we ran inference on the COCONNUT dataset. This was a much bigger dataset, and led to many potential hits. See the discussion section for how out of distribution predictions occurred

	AUROC
Dropout = 0	0.8247
Dropout = 0.1	0.8129
Dropout = 0.2	0.8392
Dropout = 0.3	0.8375
MP Depth = 0	0.8238
MP Depth = 3	0.8247
MP Depth = 5	0.8203

Seed	AUROC
0	0.8227
1	0.7518
2	0.7998
3	0.7604
4	0.8282
5	0.7876
6	0.7491
7	0.7963
8	0.796
9	0.8298